Calculus I Lesson 27

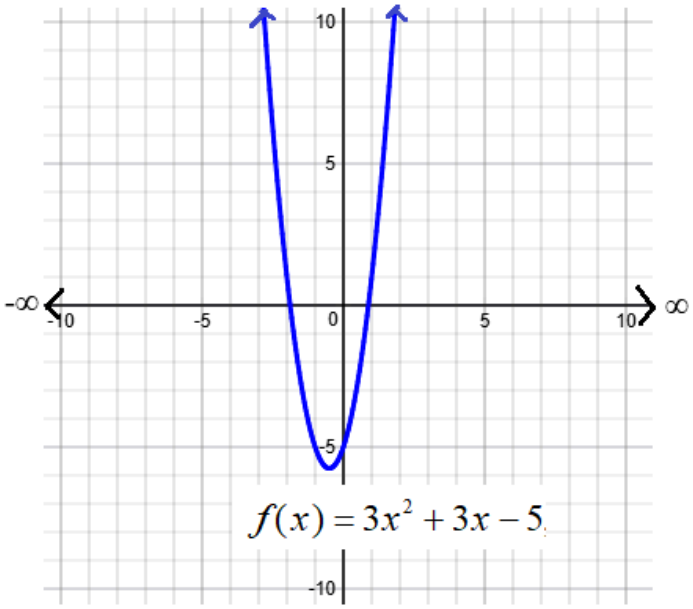
1) $f(x) = 3x^2 + 3x - 5$

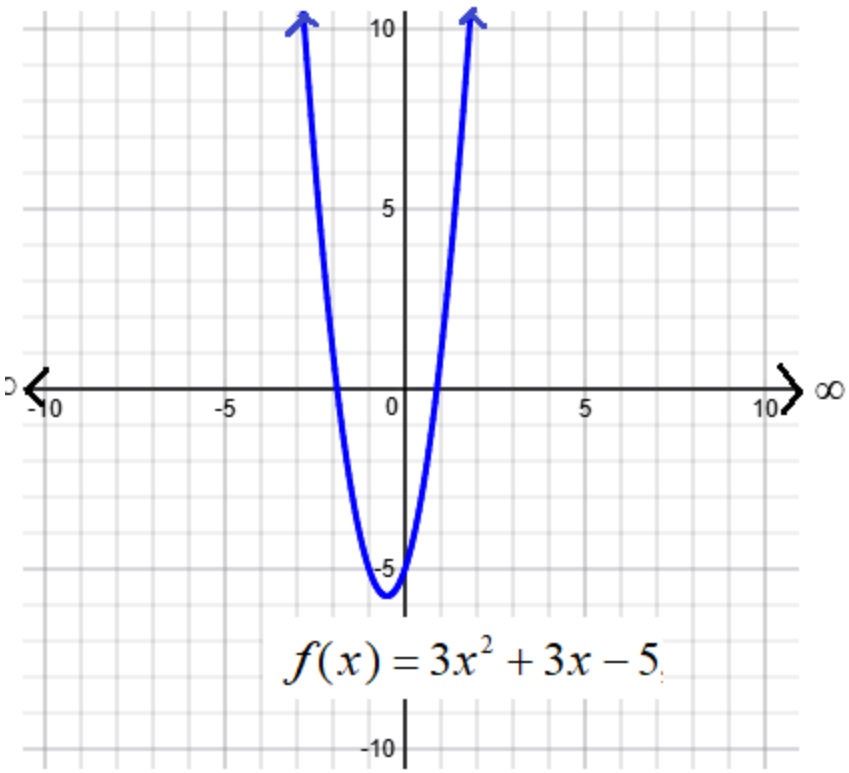
a) $f'(x) =$ ___?___

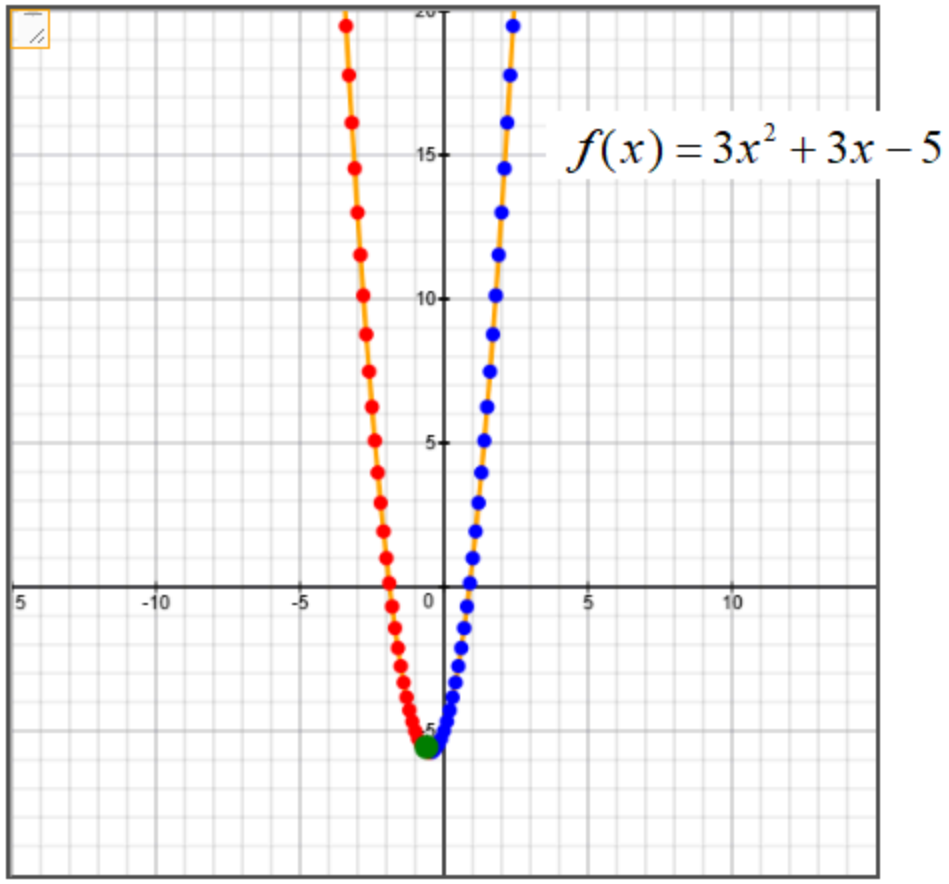
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___

c) The function $f(x) = 3x^2 + 3x - 5$ increases on the interval: ___?___

d) The function $f(x) = 3x^2 + 3x - 5$ decreases on the interval: ___?___







- $f(x) < 0$; first derivative is negative
- $f(x) > 0$; first derivative is positive
- Critical point

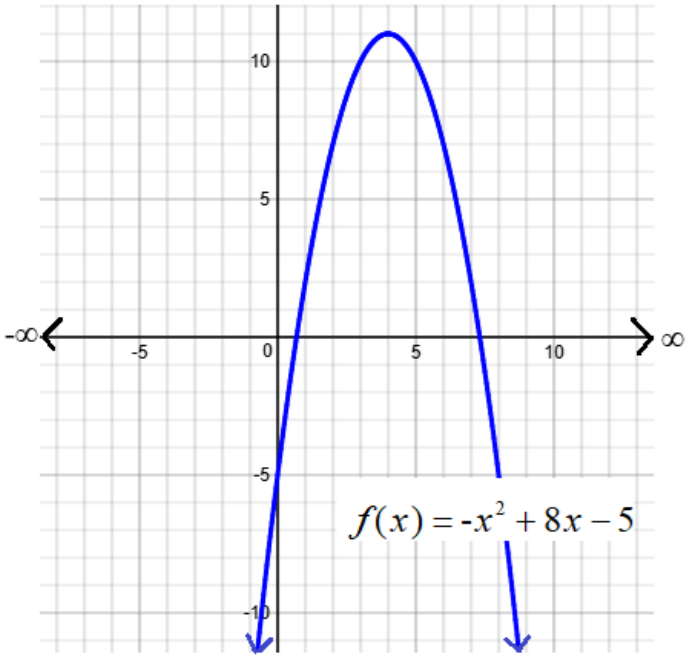
2) $f(x) = -x^2 + 8x - 5$

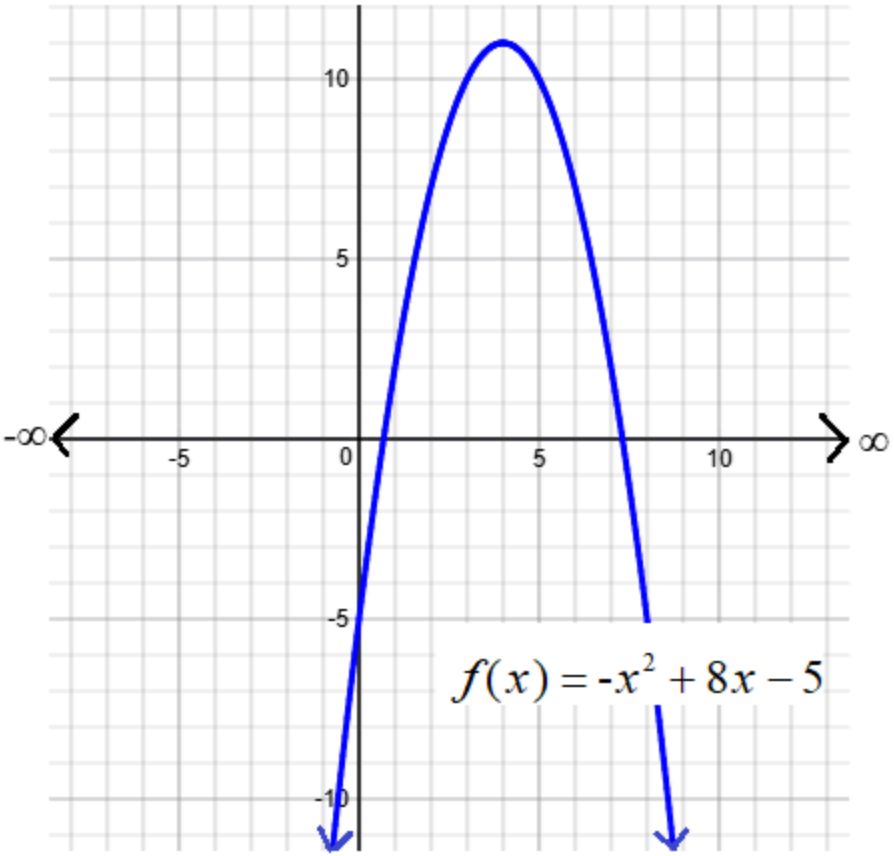
a) $f'(x) =$ ___?___

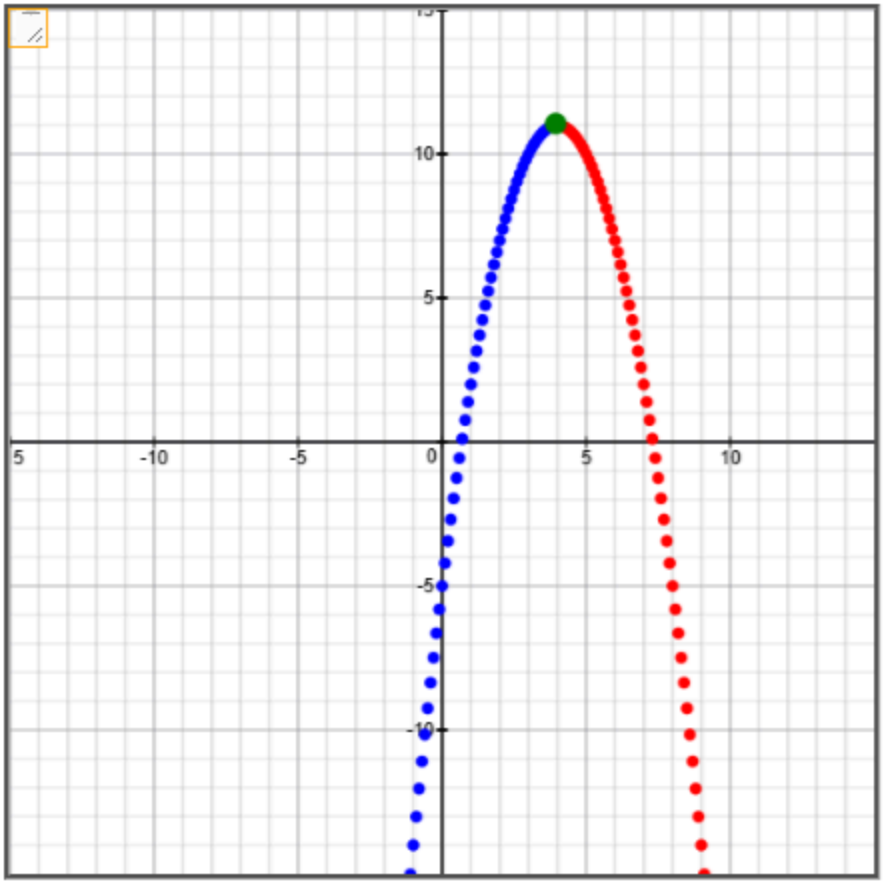
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___

c) The function $f(x) = -x^2 + 8x - 5$ increases on the interval: ___?___

d) The function $f(x) = -x^2 + 8x - 5$ decreases on the interval: ___?___







- $f(x) < 0$; first derivative is negative
- $f(x) > 0$; first derivative is positive
- Critical point

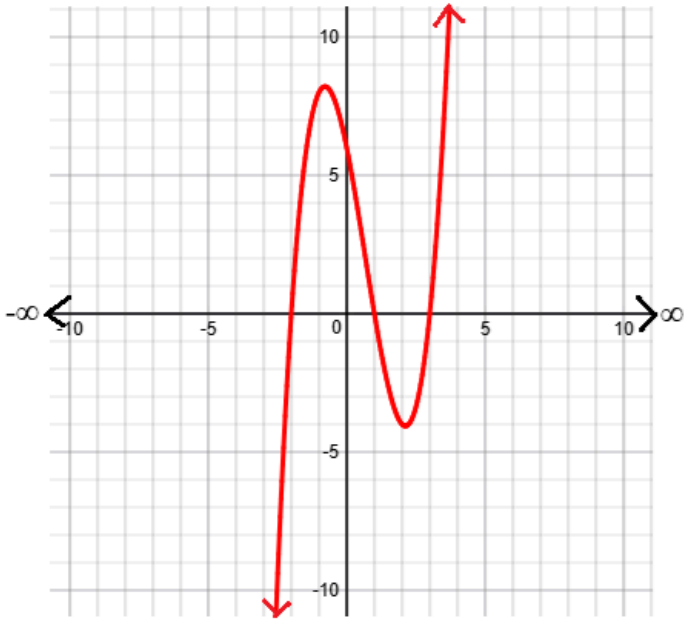
3) $f(x) = x^3 - 2x^2 - 5x + 6$

a) $f'(x) =$ ___?___

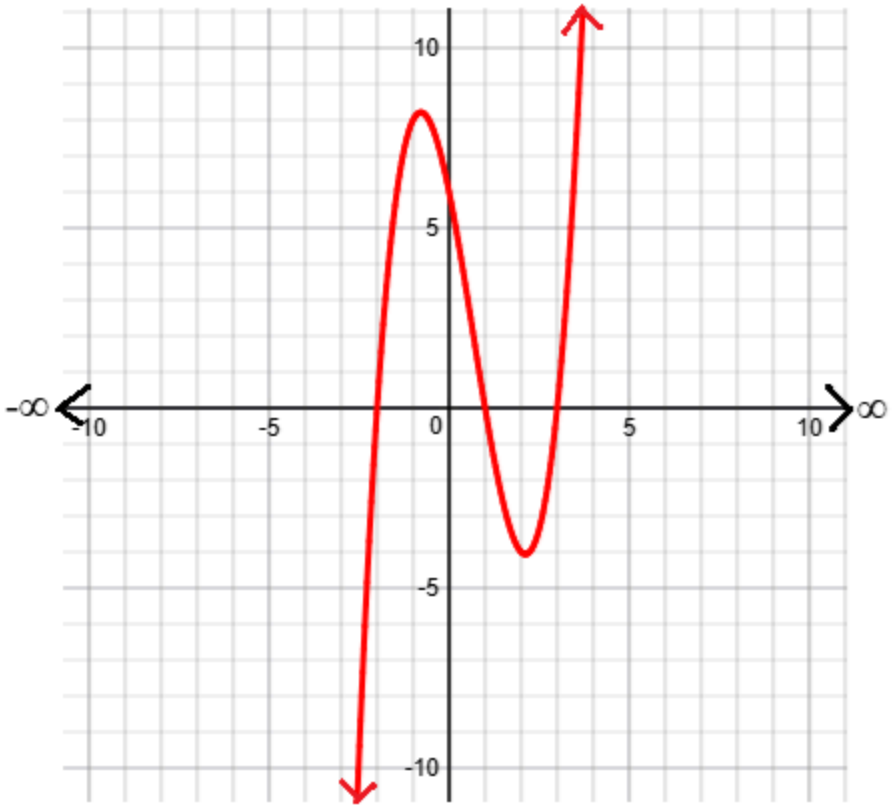
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___ and $x =$ ___?___

c) The function $f(x) = x^3 - 2x^2 - 5x + 6$ increases on the interval: ___?___

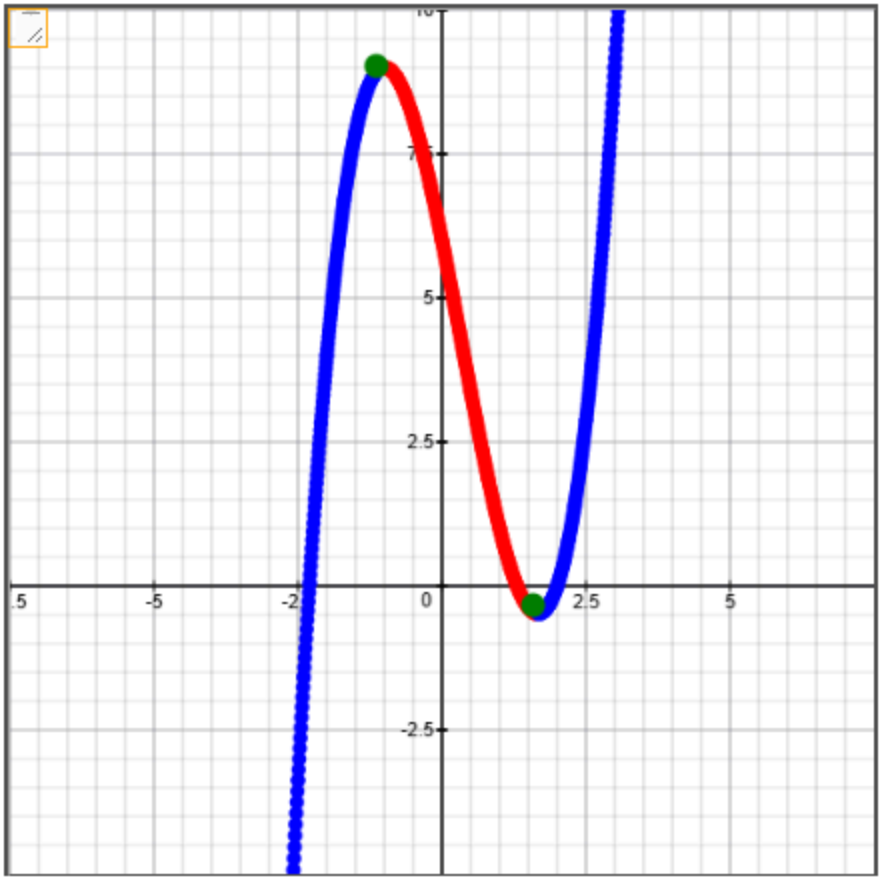
d) The function $f(x) = x^3 - 2x^2 - 5x + 6$ decreases on the interval: ___?___



$f(x) = x^3 - 2x^2 - 5x + 6$



$$f(x) = x^3 - 2x^2 - 5x + 6$$



- $f(x) < 0$; first derivative is negative
- $f(x) > 0$; first derivative is positive
- Critical point

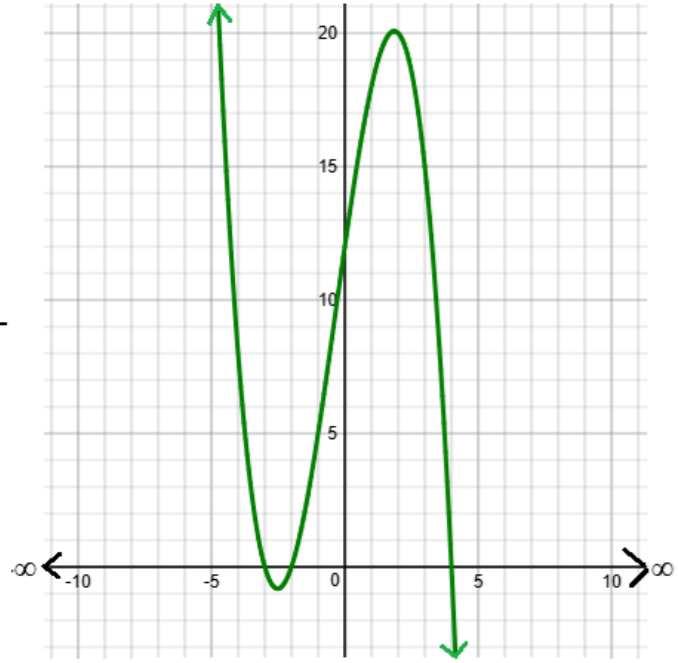
4) $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12, \quad x \in [-3, -2]$

a) $f'(x) =$ ___?___

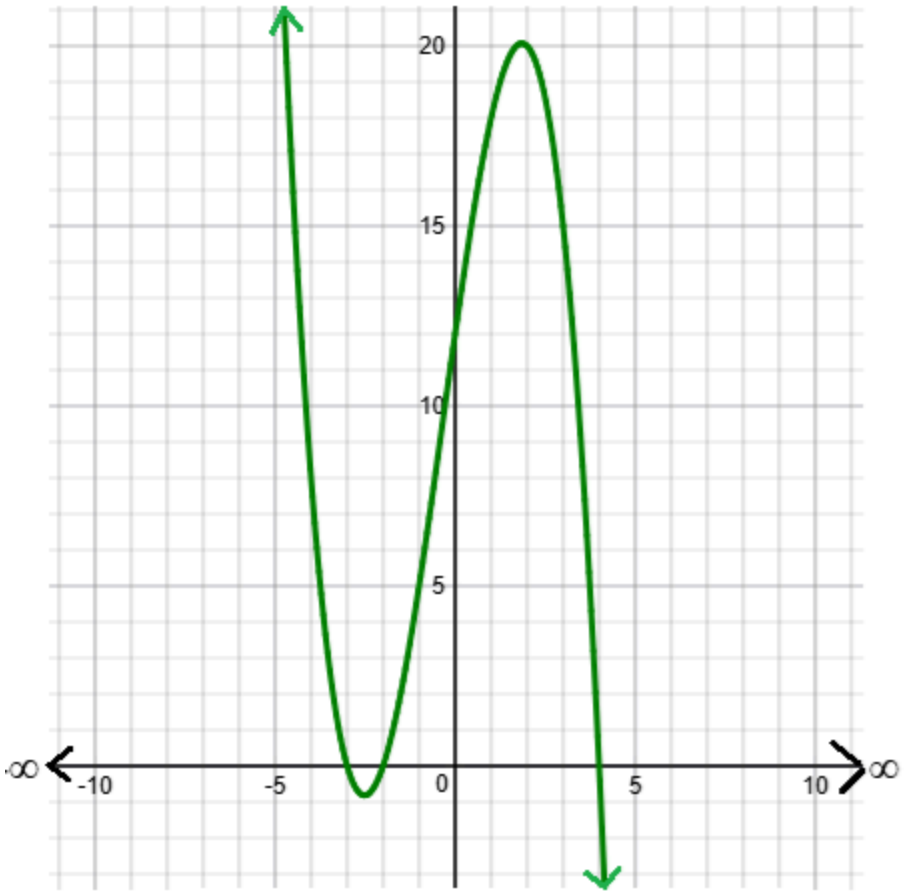
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___, $x =$ ___?___ and $x =$ ___?___

c) The function $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$ increases on the interval: ___?___

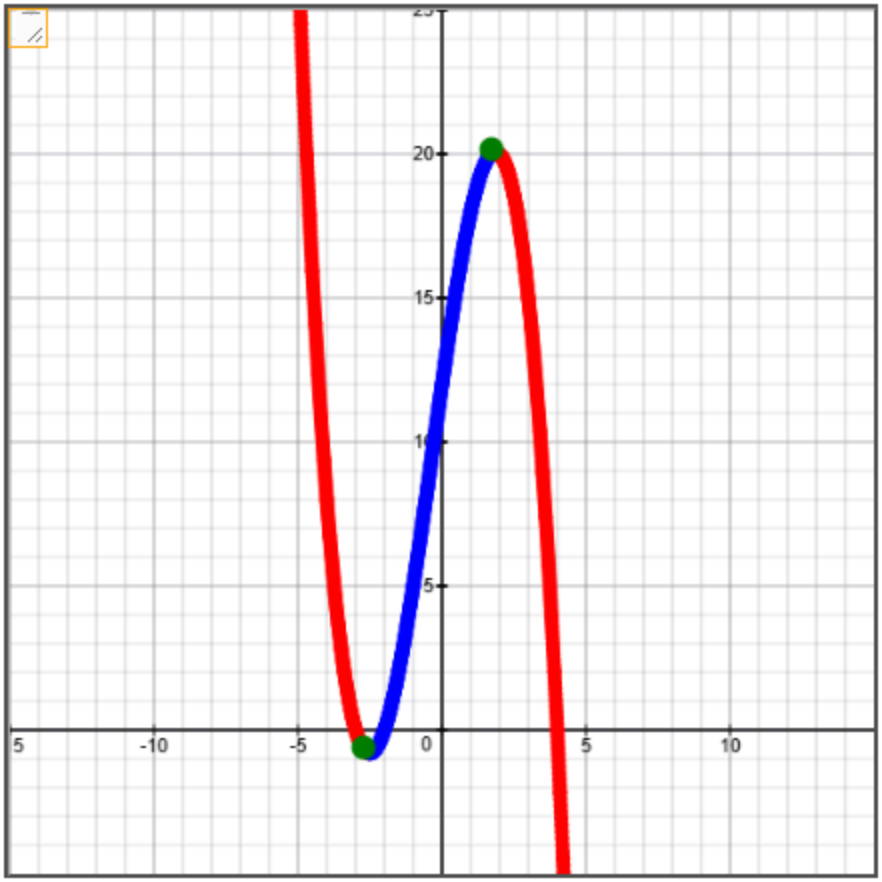
d) The function $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$ decreases on the interval: ___?___



$f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$



$$f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$$



- $f(x) < 0$; first derivative is negative
- $f(x) > 0$; first derivative is positive
- Critical point

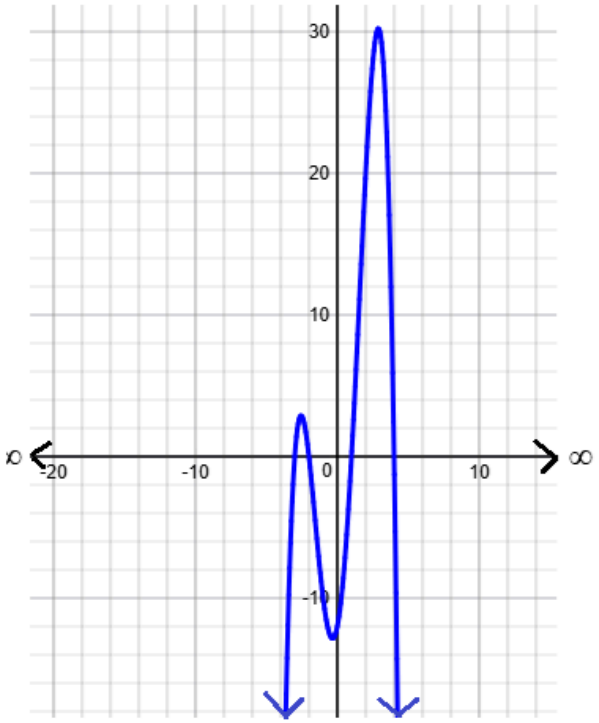
5) $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$

a) $f'(x) =$ ___?___

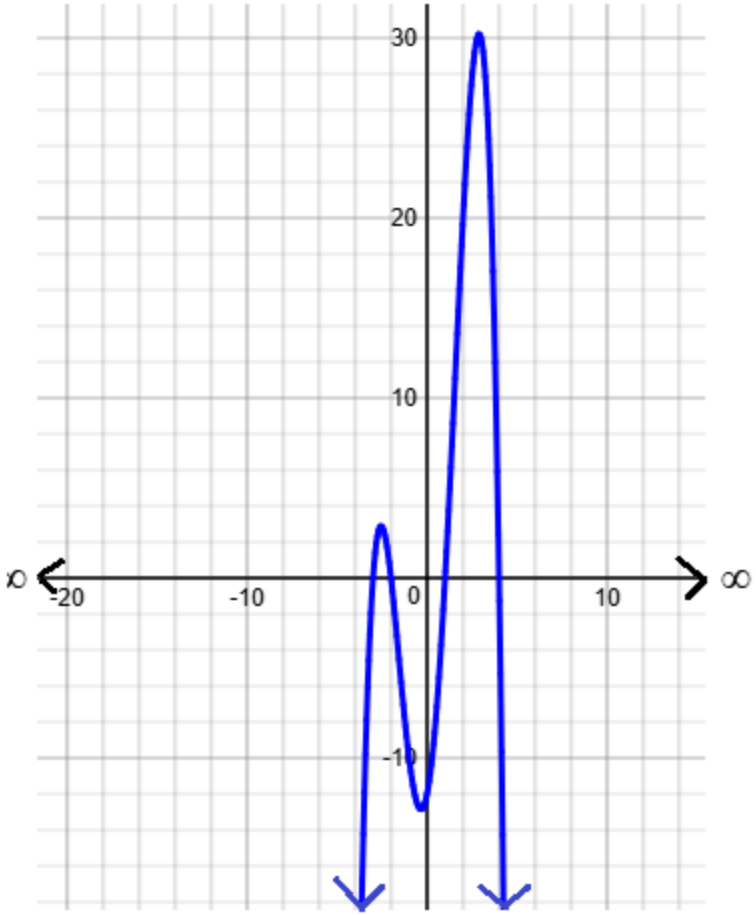
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___, $x =$ ___?___ and $x =$ ___?___

c) The function $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$ increases on the interval: ___?___

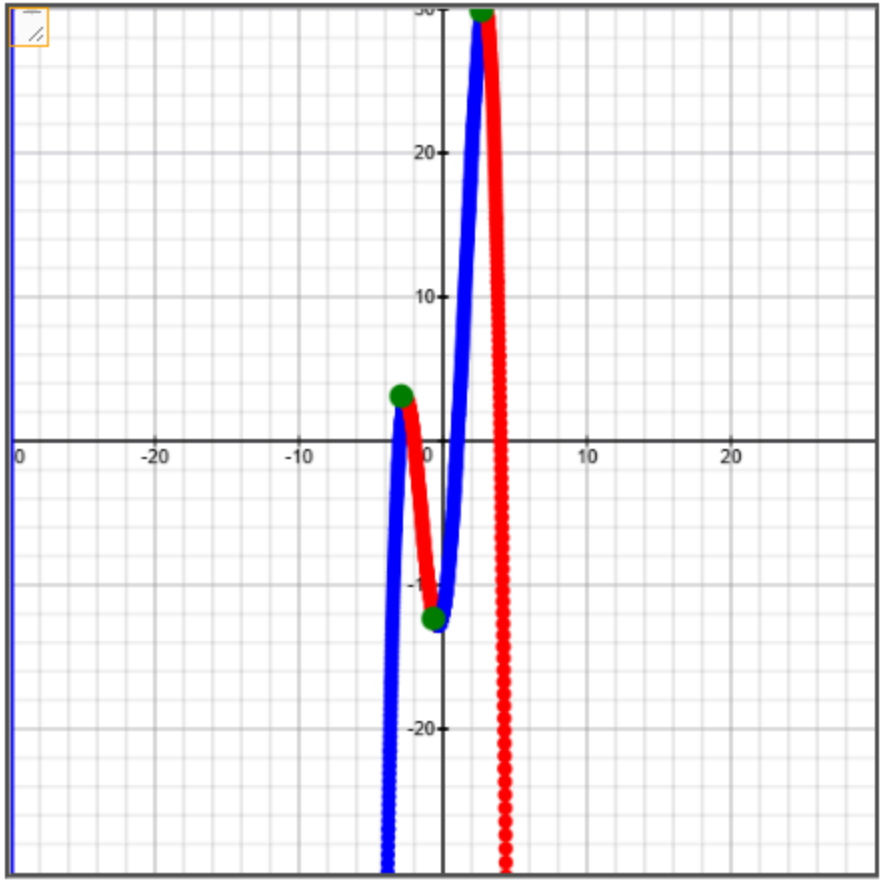
d) The function $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$ decreases on the interval: ___?___



$f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$



$$f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$$



- $f(x) < 0$; first derivative is negative
- $f(x) > 0$; first derivative is positive
- Critical point

6) $f(x) = x^{2/3} + 1$

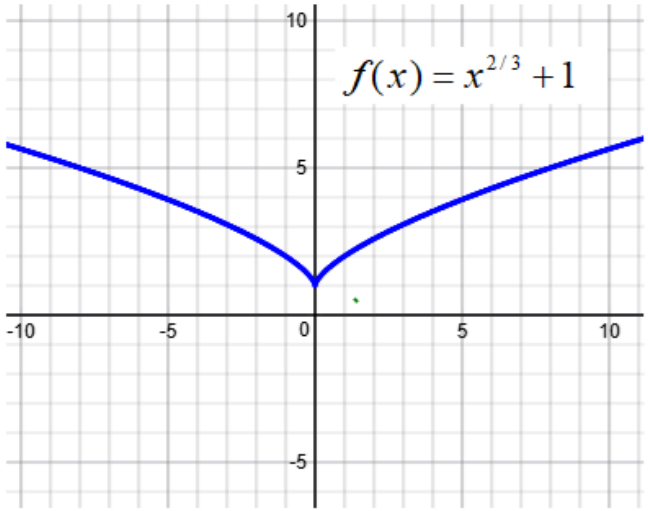
a) $f'(x) =$ ___?___

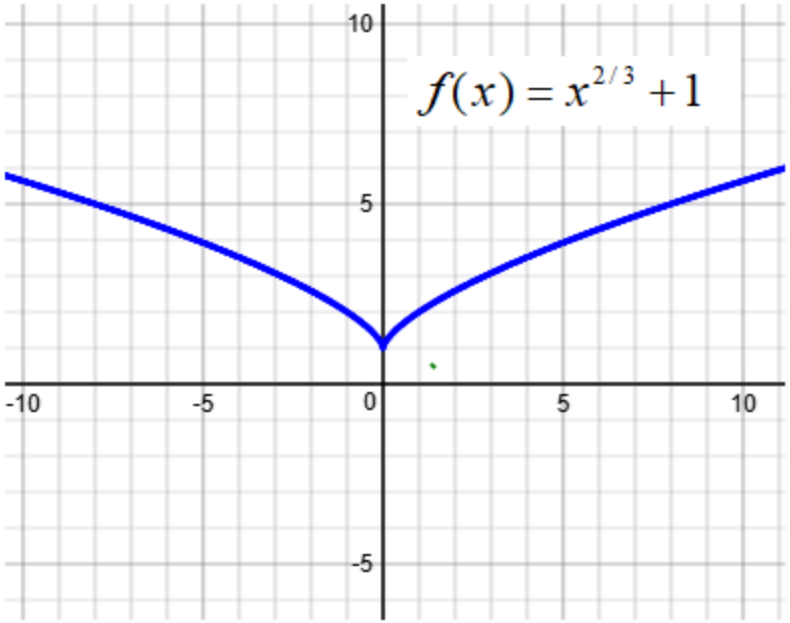
b) Set numerator of $f'(x) = 0$ and solve for x. $x =$ ___?___ (if any).

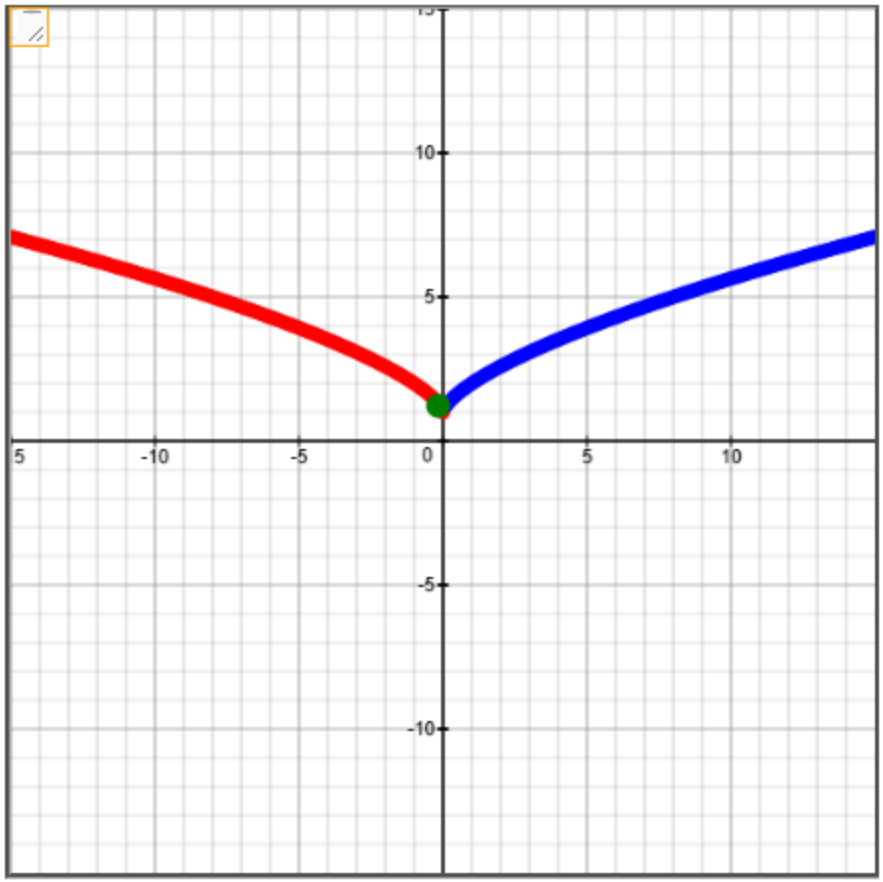
c) Set denominator of $f'(x) = 0$ and solve for x. $x =$ ___?___ (if any).

d) The function $f(x) = x^{2/3} + 1$ increases on the interval: ___?___

e) The function $f(x) = x^{2/3} + 1$ decreases on the interval: ___?___







- $f(x) < 0$; first derivative is negative
- $f(x) > 0$; first derivative is positive
- Critical point

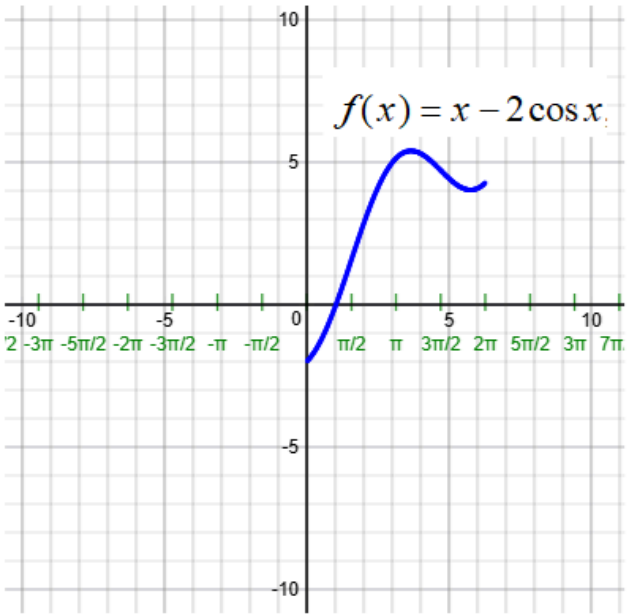
7) $f(x) = x - 2\cos x, \qquad 0 < x < 2\pi$

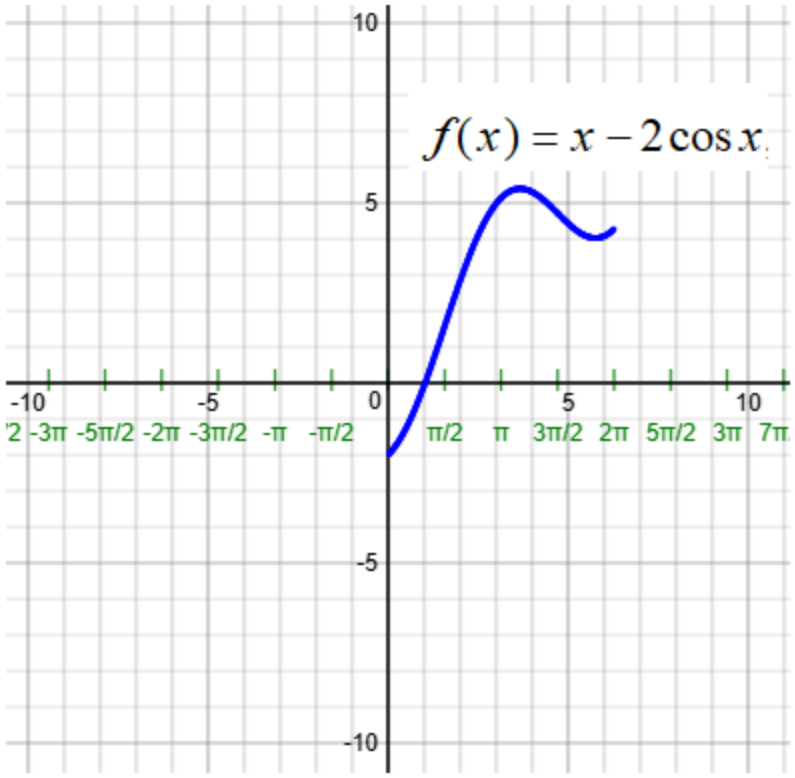
a) $f'(x) =$ ___?___

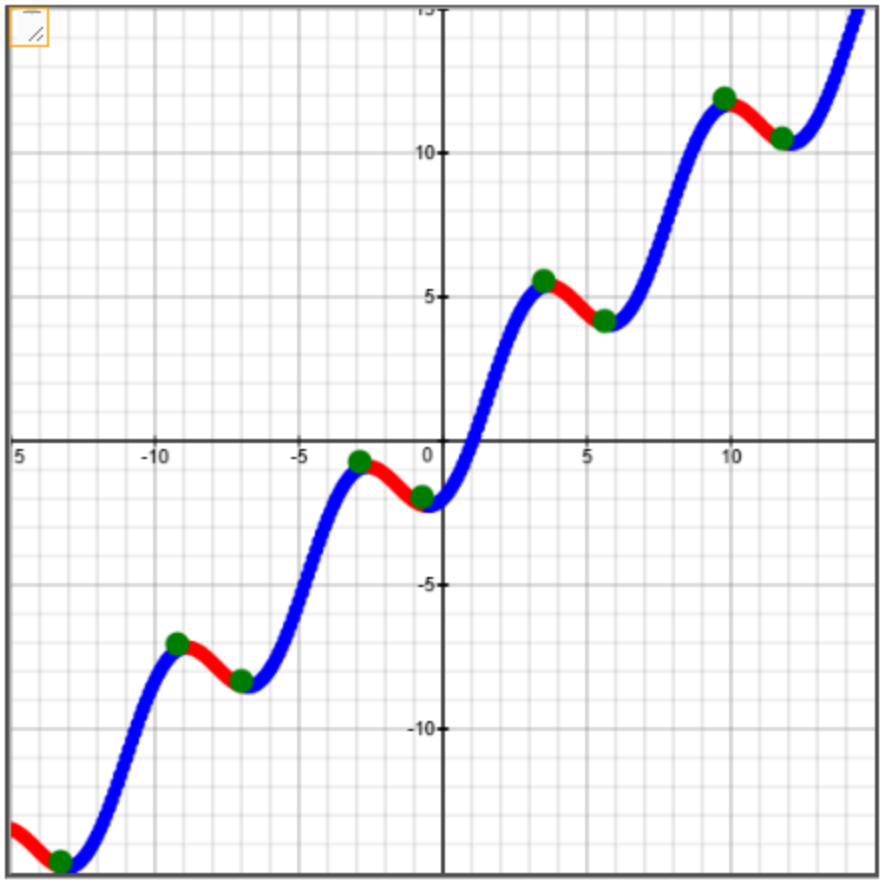
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___.

c) The function $f(x) = x - 2\cos x$ increases on the interval: ___?___

d) The function $f(x) = x - 2\cos x$ decreases on the interval: ___?___







- $f(x) < 0$; first derivative is negative
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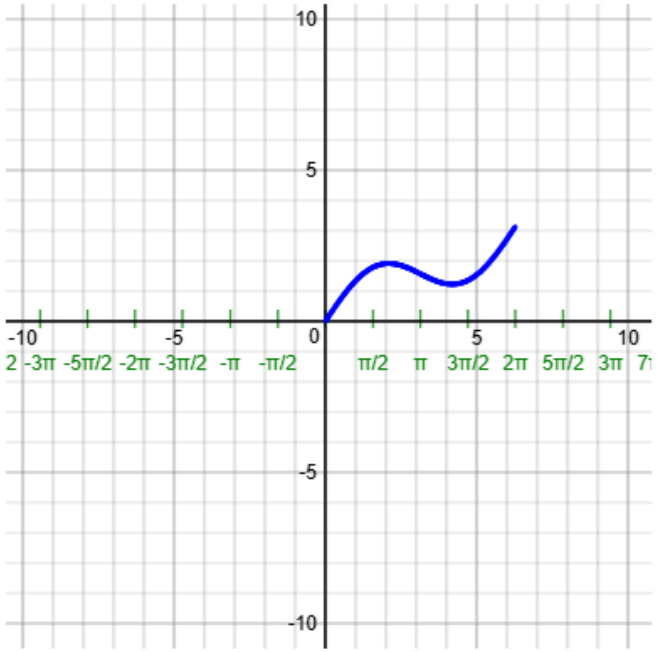
8) $f(x) = \frac{1}{2}x + \sin x, \qquad 0 \le x \le 2\pi$

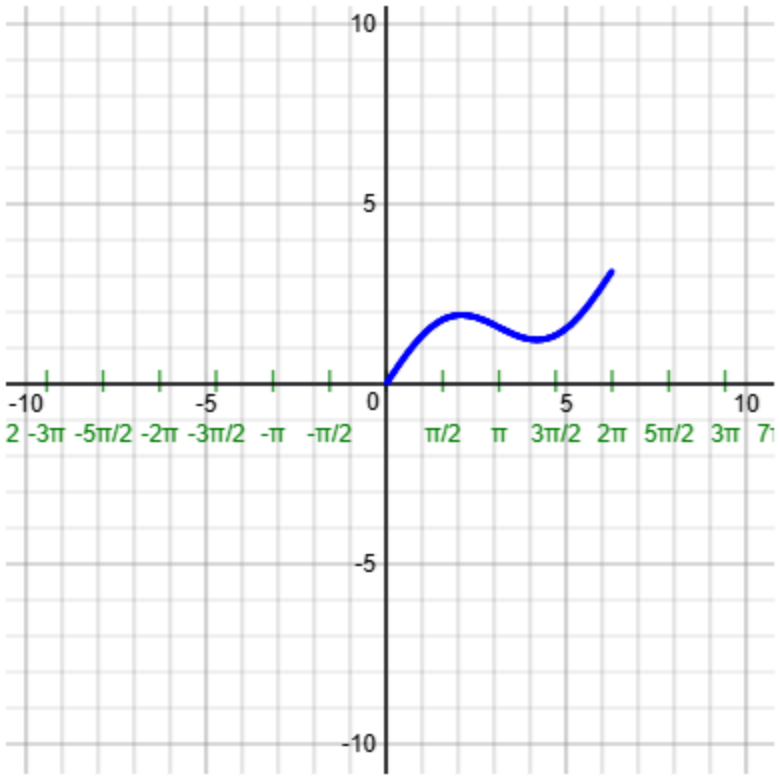
a) $f'(x) =$ ___?___

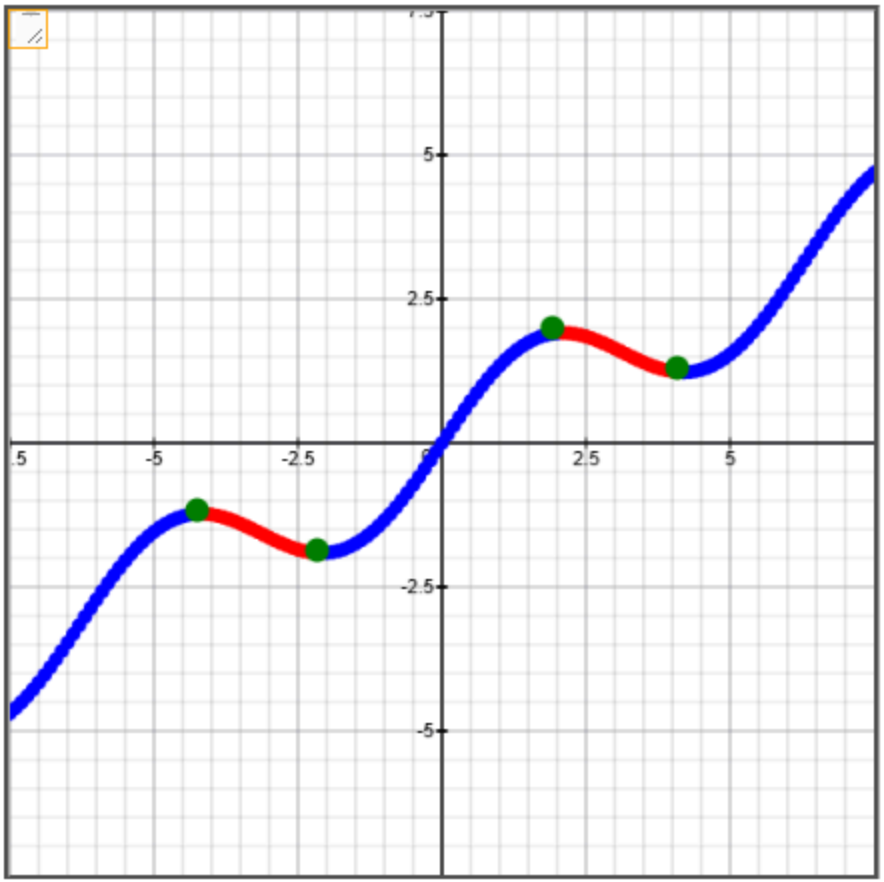
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___.

c) The function $f(x) = \frac{1}{2}x + \sin x$ increases on the interval: ___?___

d) The function $f(x) = \frac{1}{2}x + \sin x$ decreases on the interval: ___?___







- $f(x) < 0$; first derivative is negative
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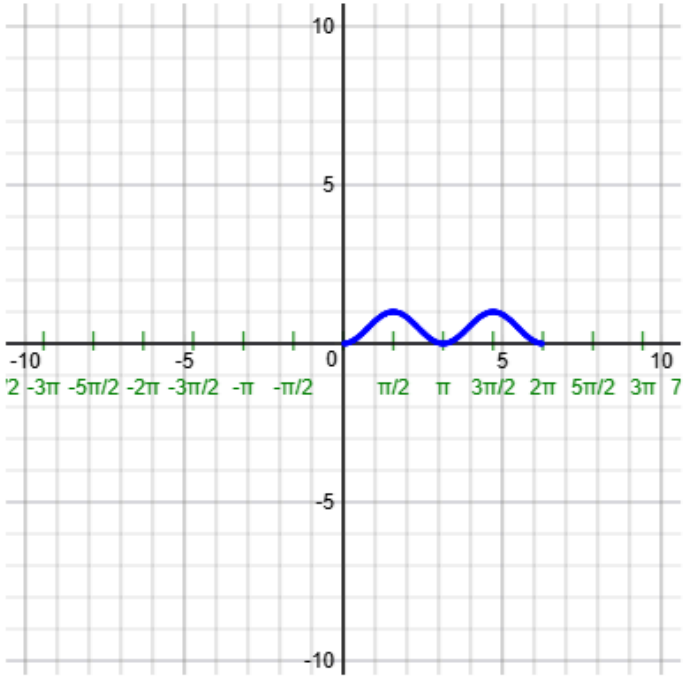
9) $f(x) = \sin^2 x = (\sin x)^2, \qquad 0 \le x \le 2\pi$

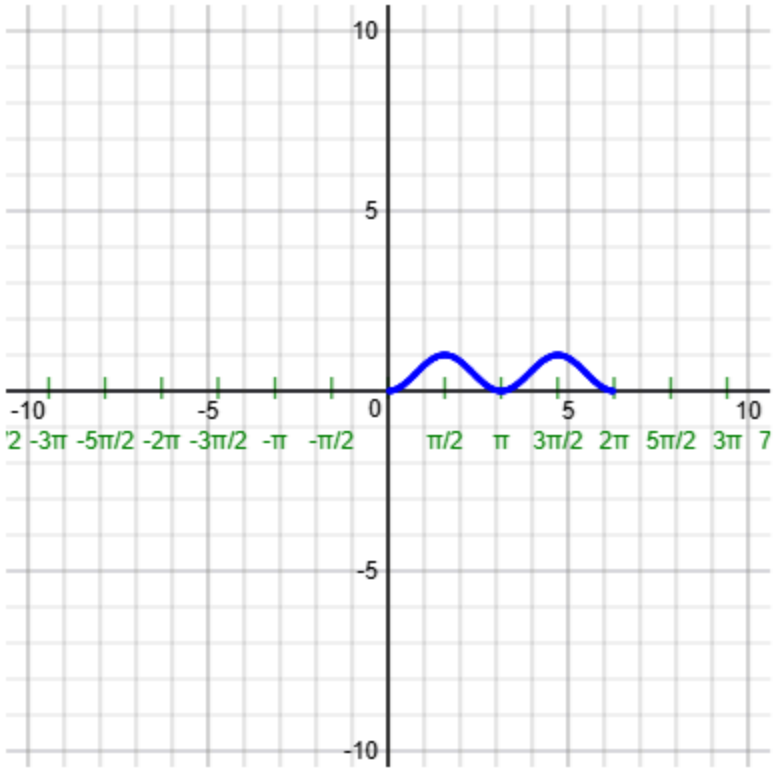
a) $f'(x) =$ ___?___

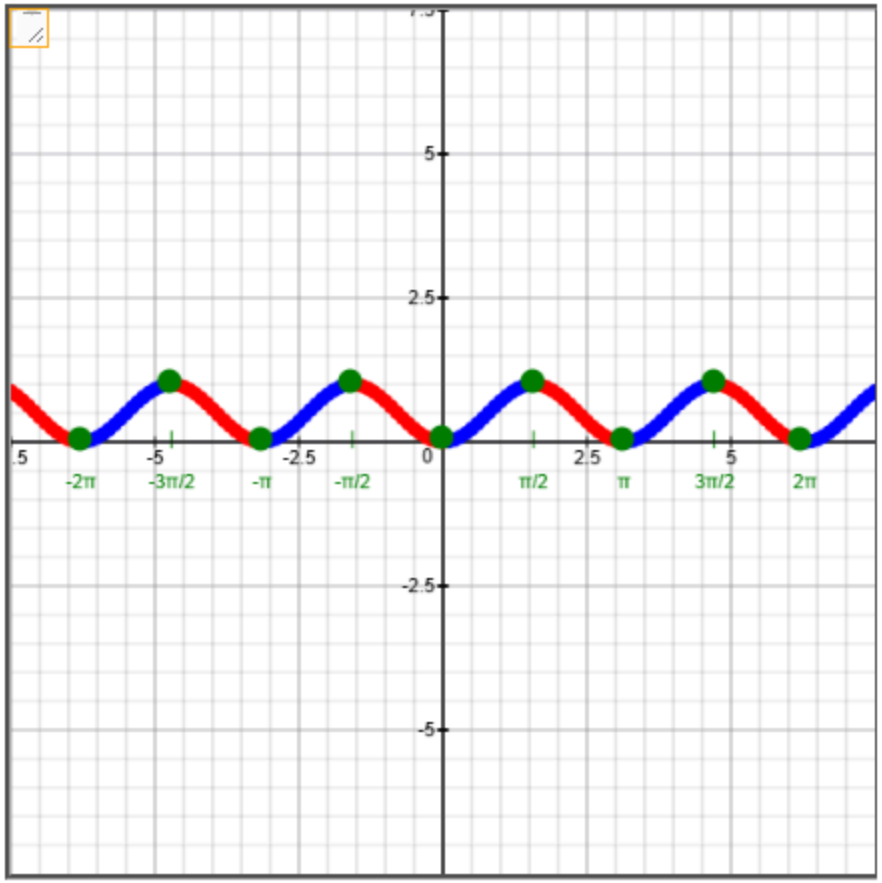
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___.

c) The function $f(x) = \sin^2 x$ increases on the interval: ___?___

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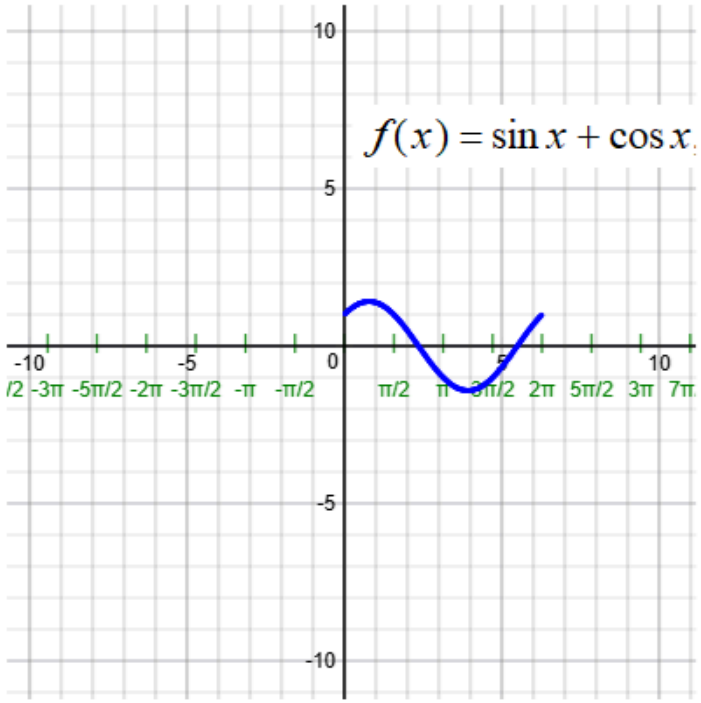
10) $f(x) = \sin x + \cos x, \qquad 0 \le x \le 2\pi$

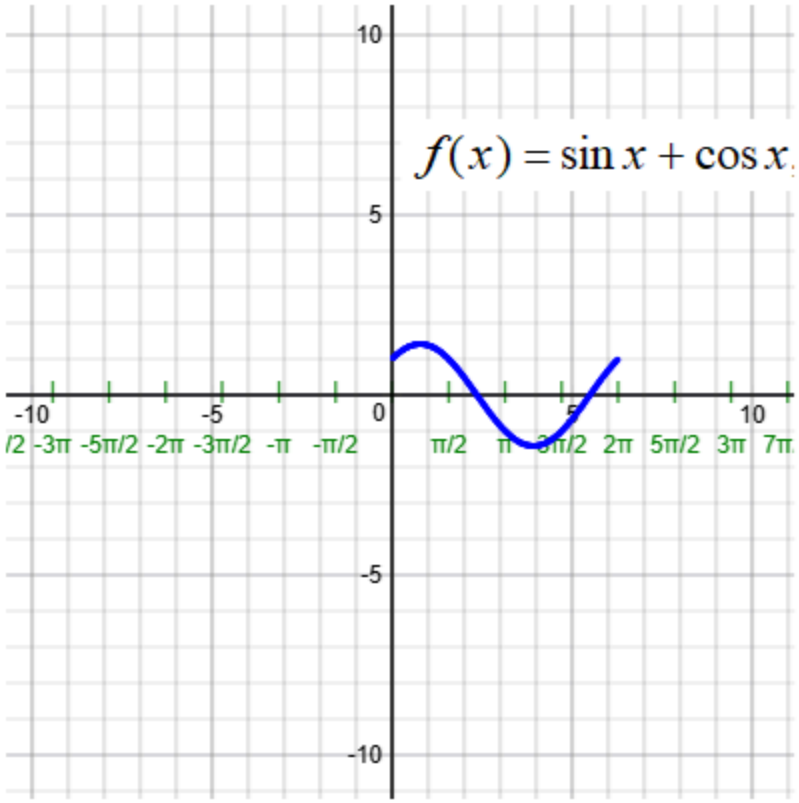
a) $f'(x) =$ ___?___

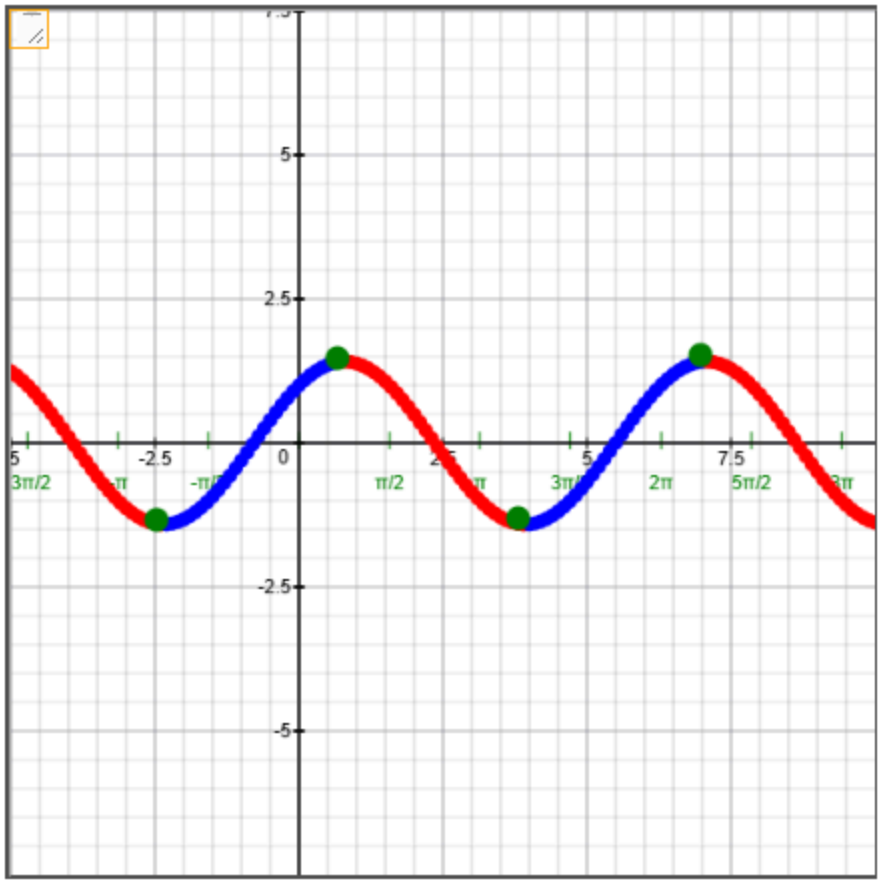
b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___.

c) The function $f(x) = \sin^2 x$ increases on the interval: ___?___

d) The function $f(x) = \sin^2 x$ decreases on the interval: ___?___







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